Daily discharge forecasting on the Potomac River and hourly inflow forecasting at Gandak Barrage

This study tested whether river flow models could forecast more accurately than carrying the latest reading forward. The two cases addressed different decisions: how much water to allocate the next day on the Potomac River in the United States, and how well recent inflow measurements could anticipate a flood crest at Gandak Barrage in India.

For the Potomac, daily discharge records from 1990 to 2025 were used to compare four neural networks and a weighted average of their predictions. Across 2,192 test days, the stacked long short-term memory network had the lowest root-mean-square error. Combining the models gave the smallest average absolute error: the combined forecast differed from the observed flow by 37.79 m³/s on average, counting both overestimates and underestimates. A simulation then used these forecasts to allocate water under hypothetical demand. When allocations were based on the mean forecast, they exceeded available supply on 63.5% of days. Planning instead for a low-flow estimate (the fifth percentile) reduced that figure to 1.5%, but more demand went unmet. The basin prototype also tracks water balance and includes satellite methods for mapping rice, surface wetness, floods, slope changes and turbidity. These methods await field validation. A separate review examined phosphorus data for constructed wetlands.

At Gandak, regression models used recent inflow to predict each of the next 72 hours. In tests from 16 September to 27 October 2026, using the latest reading gave a smaller average error than regression at every lead time. On 27 September, the model predicted that inflow would peak at 12,519 m³/s over the next three days. Just 14 hours later, measured inflow reached 15,396 m³/s. Recent inflow alone gave little warning of the size of this crest.